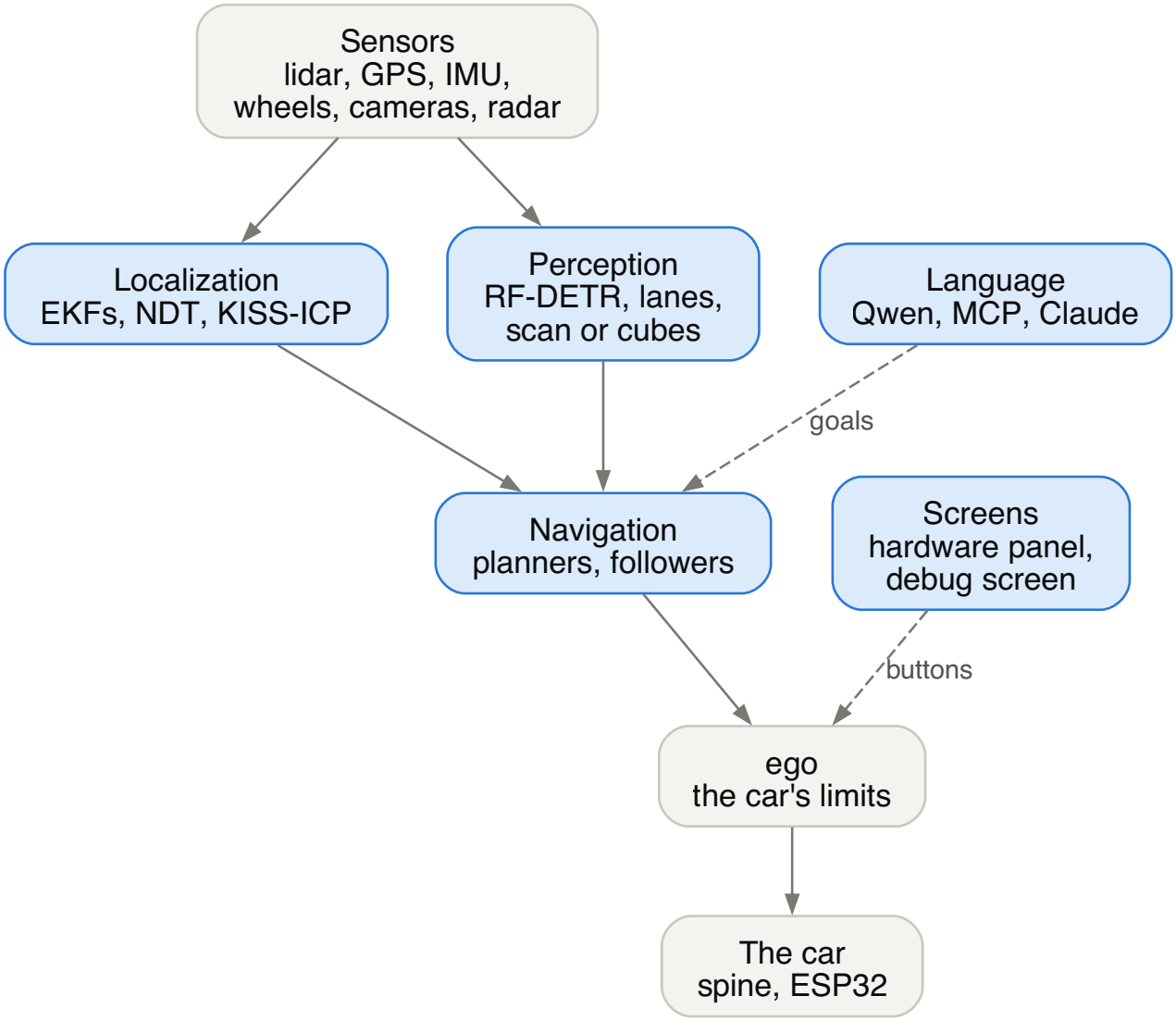
CARLETON UNIVERSITY · AUTONOMOUS VEHICLE CAPSTONE

What's new: 22 October to 1 November

Everything the last few pull requests added (#13 to #19, and the branch after), by area; then the best version to take to the car, what's left for CARLA, and which Ubuntu each computer should run. Every feature, with who built it: Features. How to switch each trial on: Switches.

The short version

- **Screens:** a hardware panel with simple buttons, and a debug screen that shows what the car sees in 3D.
- **Where am I:** the lidar's point cloud now feeds NDT (a map match) and KISS-ICP (how far it moved), beside the two EKFs.
- **What's around:** RF-DETR (not YOLO) for objects, OpenCV for lanes, the lidar as a flat scan or as cubes.
- **Getting there:** five planners, six path followers, reversing and the U-turn.
- **Talking to it:** a local Qwen model with our own harness, and MCP so Claude can drive the sim.
- Everything here has run in Gazebo; **none of it has driven the real car yet.**



Blue: built in the last few pull requests. Grey: there before.

Status: sim = works in Gazebo; **built** = built and tested, not yet in a full run.

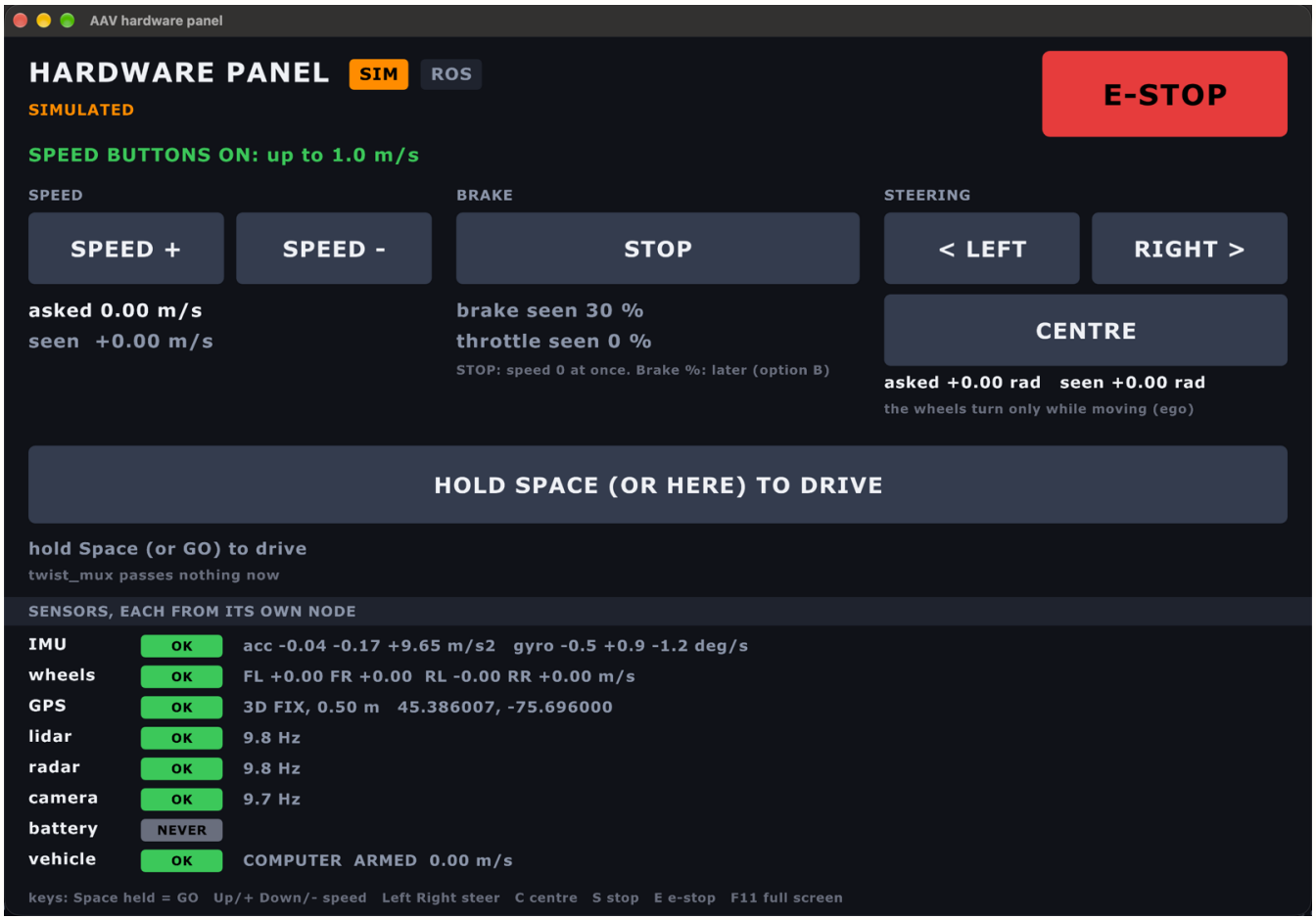
Inside

Hardware interface: the two screens	3	Talking to the car	15
Localization: where am I?	5	The best version: what goes on the car	16
Perception: what's around?	8	CARLA: what's left to work with our graph	16
Navigation in 2D and 3D: the lidar for Nav2	9	The computers: which Ubuntu	17
Navigation: the route and the steering	11	Tools to find all this again	18

Hardware interface: the two screens

The hardware panel (`pixi run hw-panel`)

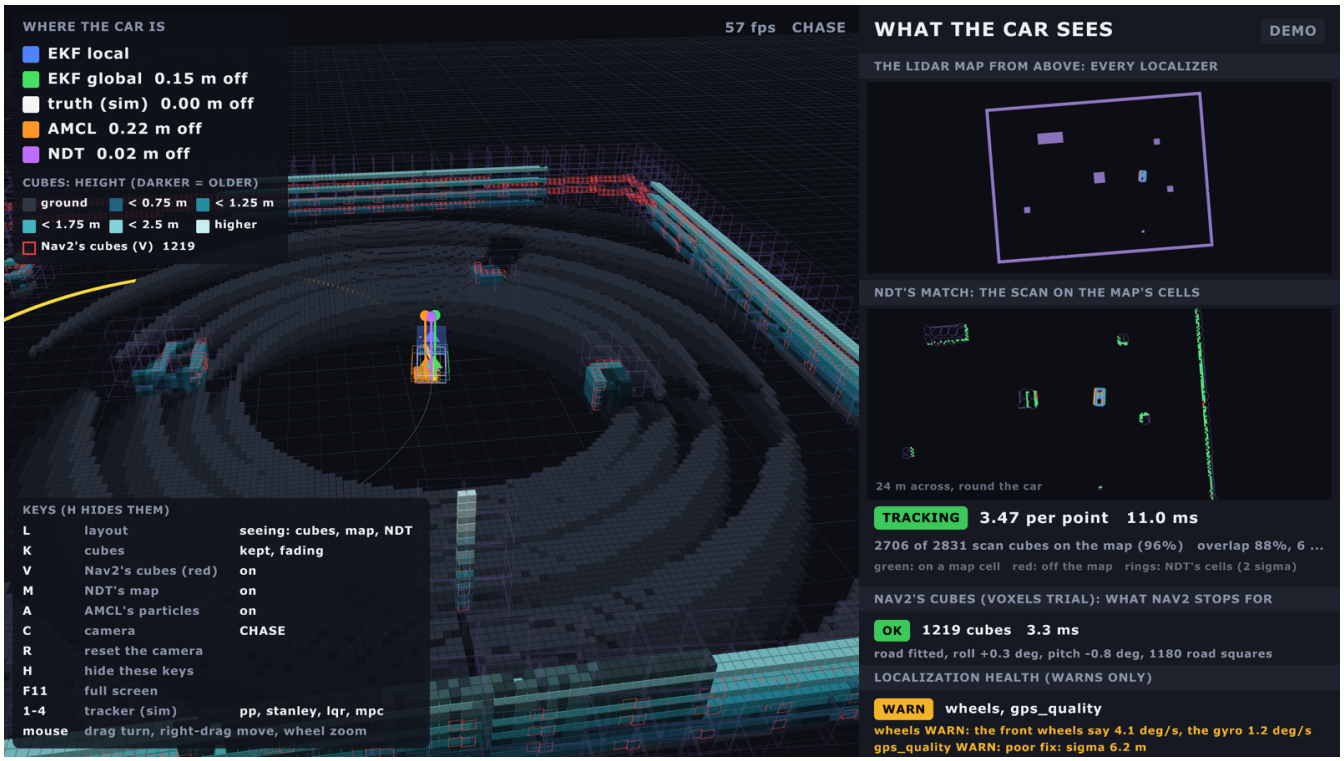
Every sensor's number from its own node, and very simple buttons: speed up and down, left, right, centre, stop. The car moves only while you **hold** Space (or GO).



What	Why	Status
OK / STALE / NEVER for each sensor	see at a glance what's missing	sim
Hold to drive, through twist_mux's slot 45	a person always outranks it; let go and it stops	sim
SIM and BENCH only	not on the real car until its own test (TODO 49b)	built

The debug screen (`pixi run debugdemo`)

Left: the lidar as cubes in 3D, and where each localizer thinks the car is. Right: the lidar map from above, and NDT's match. New on 1 Nov: **Nav2's cubes** in red (key V), and **localization health**.



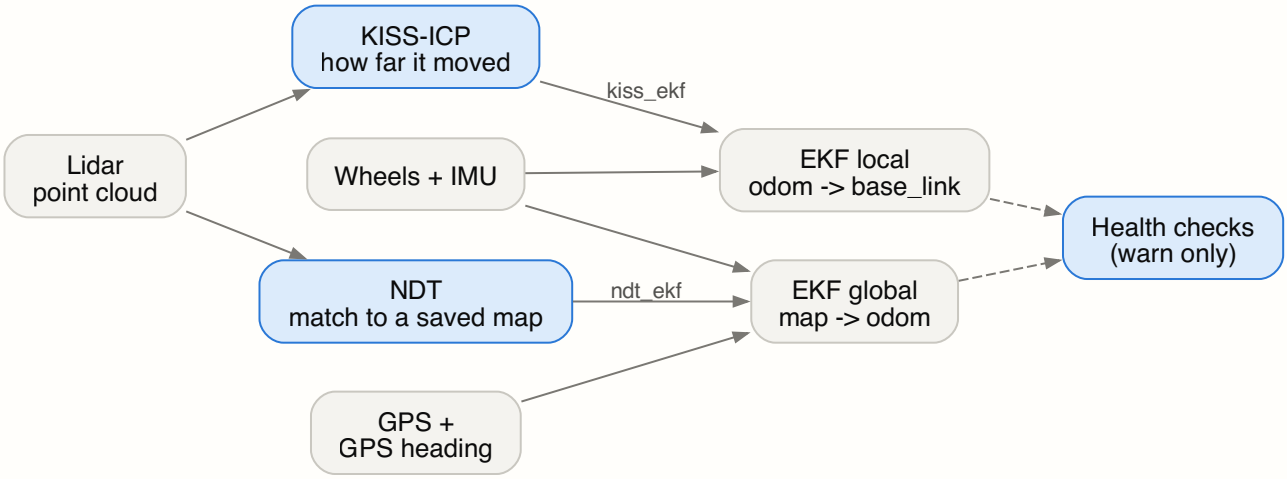
Colour	Who says where the car is
blue	EKF local (wheels, IMU)
green	EKF global (plus GPS)
purple	NDT
orange	AMCL
white	Gazebo's truth (sim only)

The firmware behind it

What	Who	Status
The spine's state machine, self-test at boot, hardware test mode	Arsh	built
The ESP32's Rust firmware: brakes 111-120 ms after the Orin's heartbeat stops	Ben	built (QEMU)

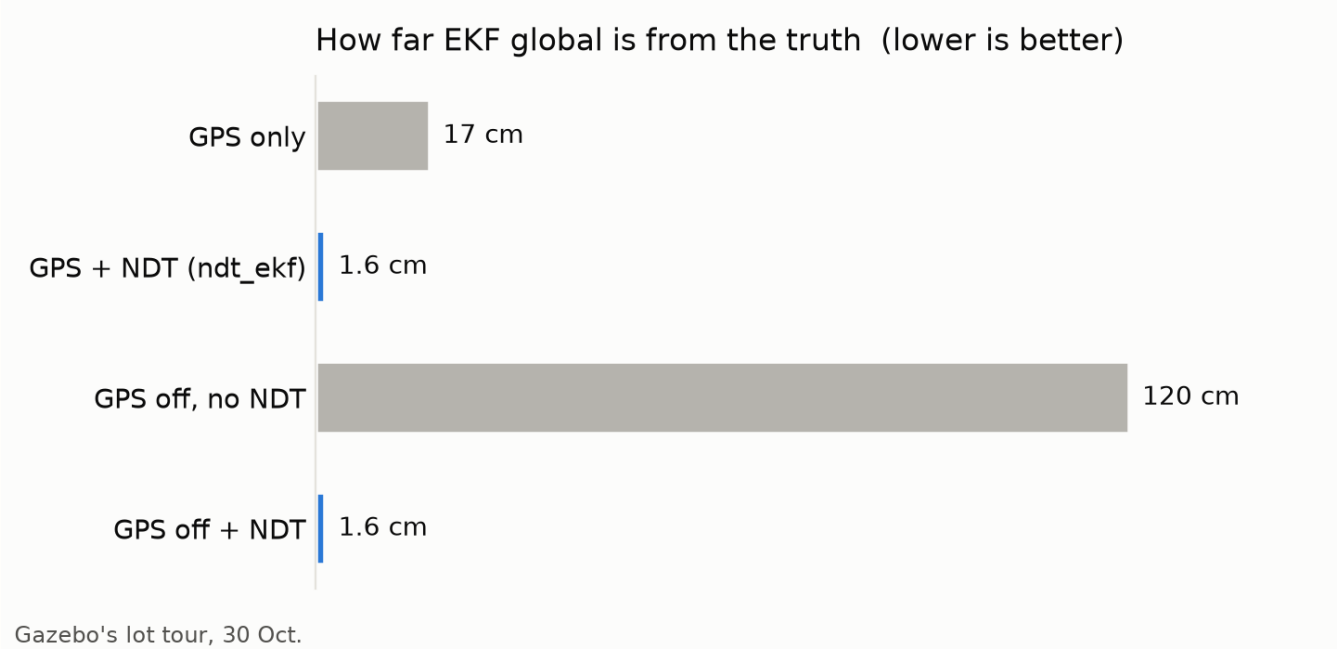
What	Who	Status
The preflight on stands (`pixi run preflight`)	Ben	sim
Geofence and stop hold (new twist_mux locks)	Ben	sim

Localization: where am I?

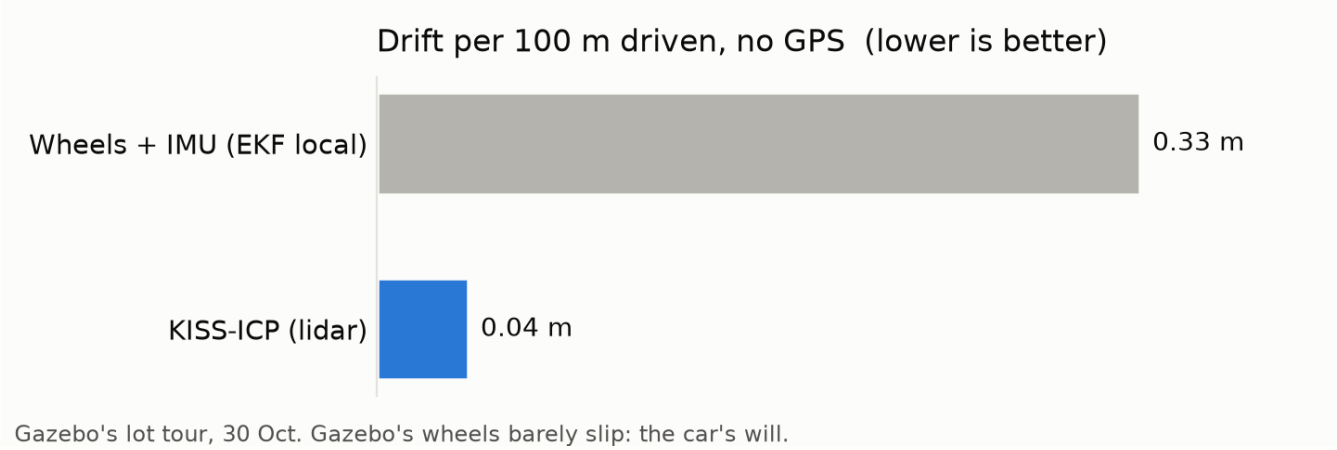


Piece	What it does	Switch	Result in Gazebo	Status
EKF local and global	the default: wheels + IMU, then GPS	always	0.2 m off the truth	sim
NDT (ours, Rust)	matches the lidar to a saved map	`ndt:=true`	1-3 cm off, 2 ms a match	sim
NDT into EKF global	uses that match to correct the map position	`ndt_ekf:=true`	17 cm → **1.6 cm**; holds with the GPS off	sim
KISS-ICP (stock, MIT)	how far the car moved, from lidar alone	`kiss_icp:=true`	0.04 m drift per 100 m (wheels: 0.33)	sim
KISS-ICP into EKF local	its speed and turn rate into the EKF	`kiss_ekf:=true`	not a clear win yet: off by default	sim
AMCL (stock)	a particle filter on a 2D map	`amcl:=true`	lost itself: the EKF wins	built

Piece	What it does	Switch	Result in Gazebo	Status
Health checks (Betsi)	warns on gyro bias, a tilted IMU, bad GPS, wrong frames	`health:=true`	on the debug screen, in `pixi run check`	sim



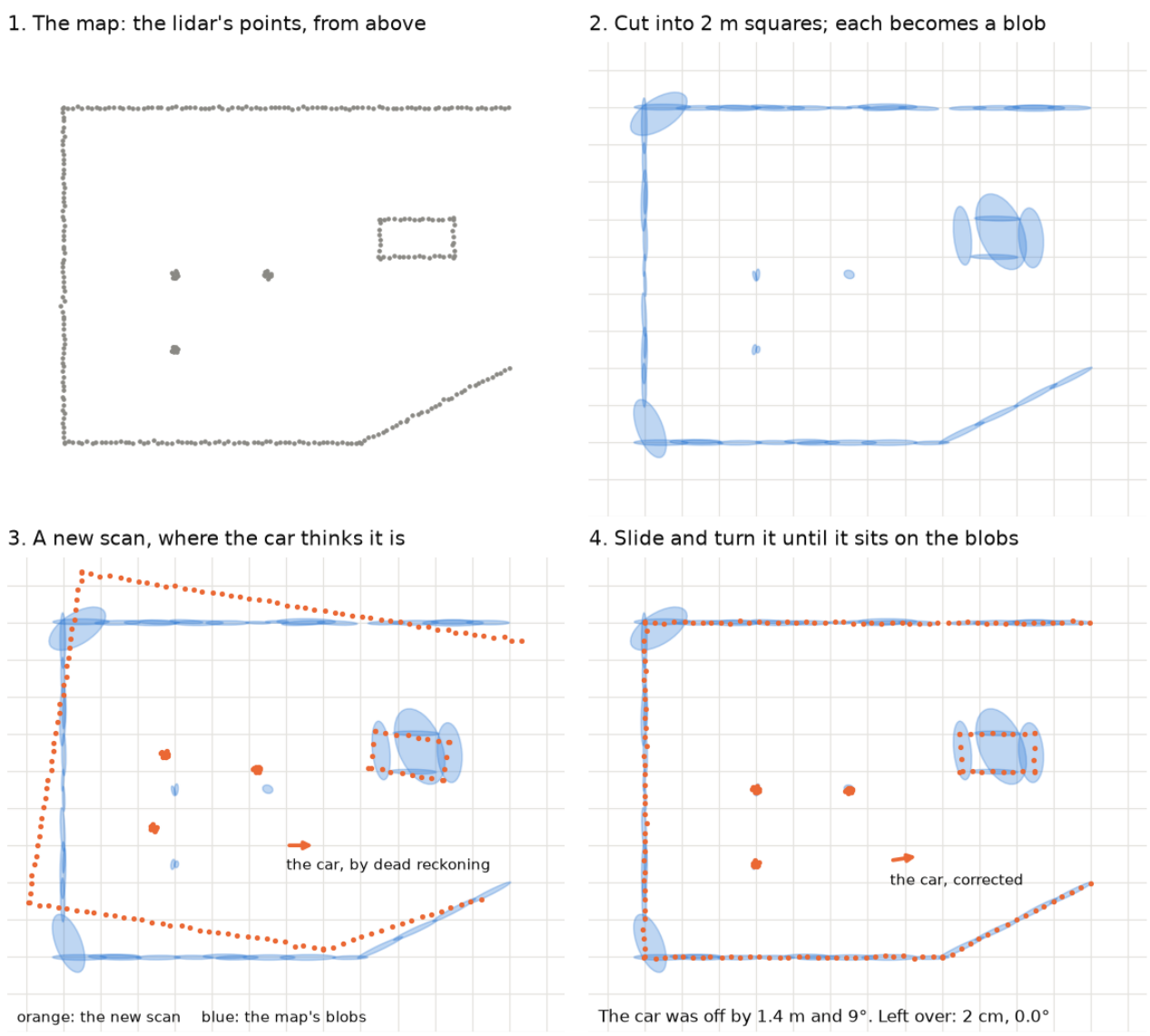
Gazebo's lot tour, 30 Oct.



Gazebo's lot tour, 30 Oct. Gazebo's wheels barely slip: the car's will.

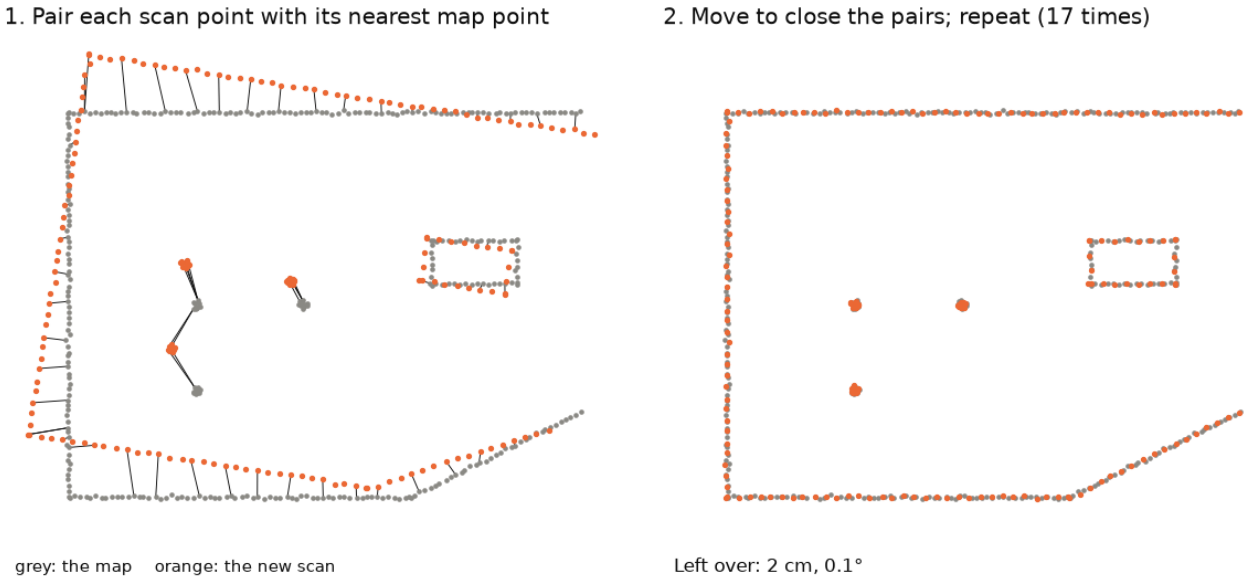
NDT, in one picture

The map becomes blobs; a new scan slides and turns until it sits on them.



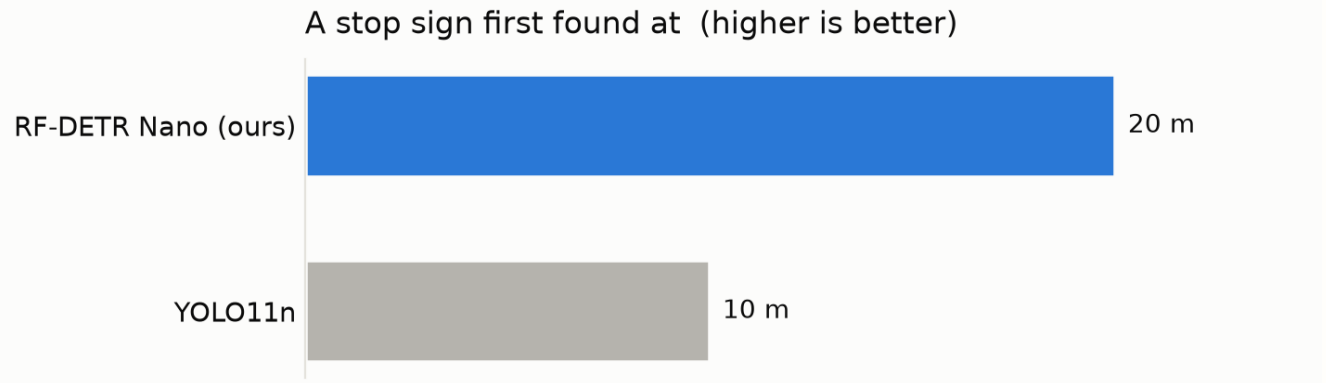
ICP, in one picture

Each scan point pairs with its nearest map point; move to close the pairs; repeat.

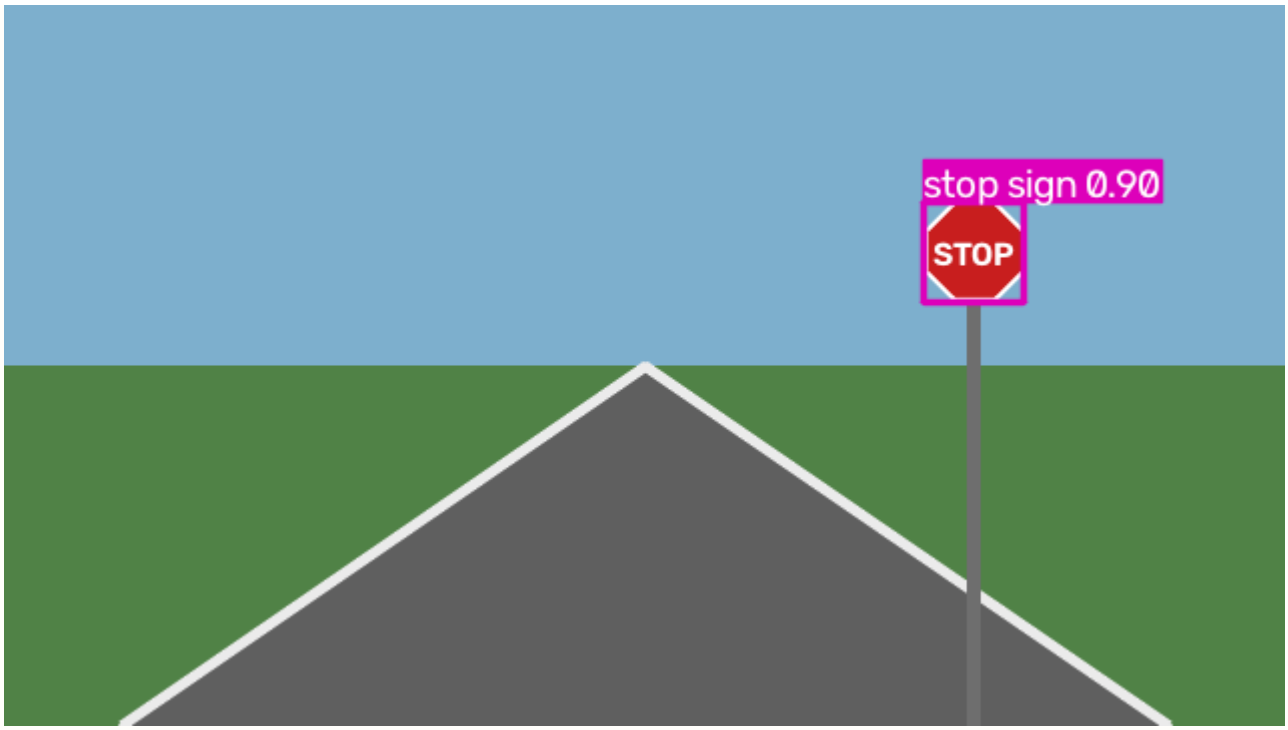


Perception: what's around?

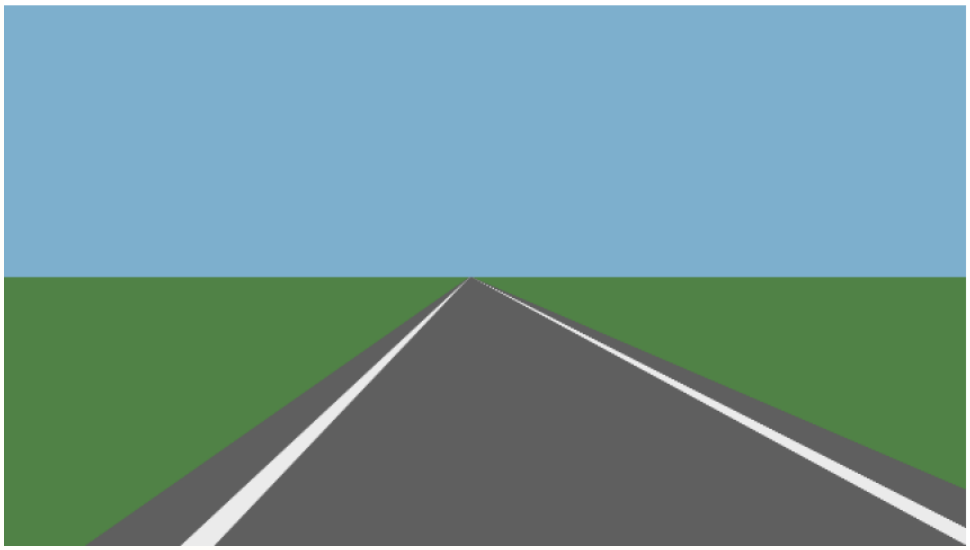
Piece	What it does	Result	Status
RF-DETR Nano (not YOLO; Apache-2.0)	finds objects and stop signs	a stop sign from 20 m (YOLO11n: 10 m), no false finds	sim
A second detector on a Mac (Core ML)	a second opinion over WiFi	21 ms a picture on a MacBook	sim
Lane finder (classic OpenCV, no model)	finds the lane from the camera	0.78 ms a picture	sim
Radar clustering (Rust)	radar points into objects	the same answers as the C++, bit for bit	sim



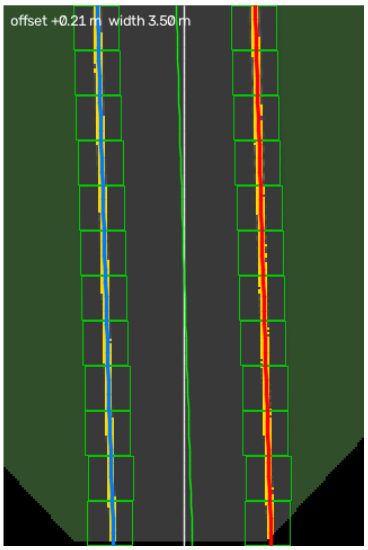
Gazebo road world, 20 Oct. RF-DETR: Apache-2.0, no false finds.



The bench camera: the car 0.3 m left of the centre, turning back



lane_finder's bird's-eye view

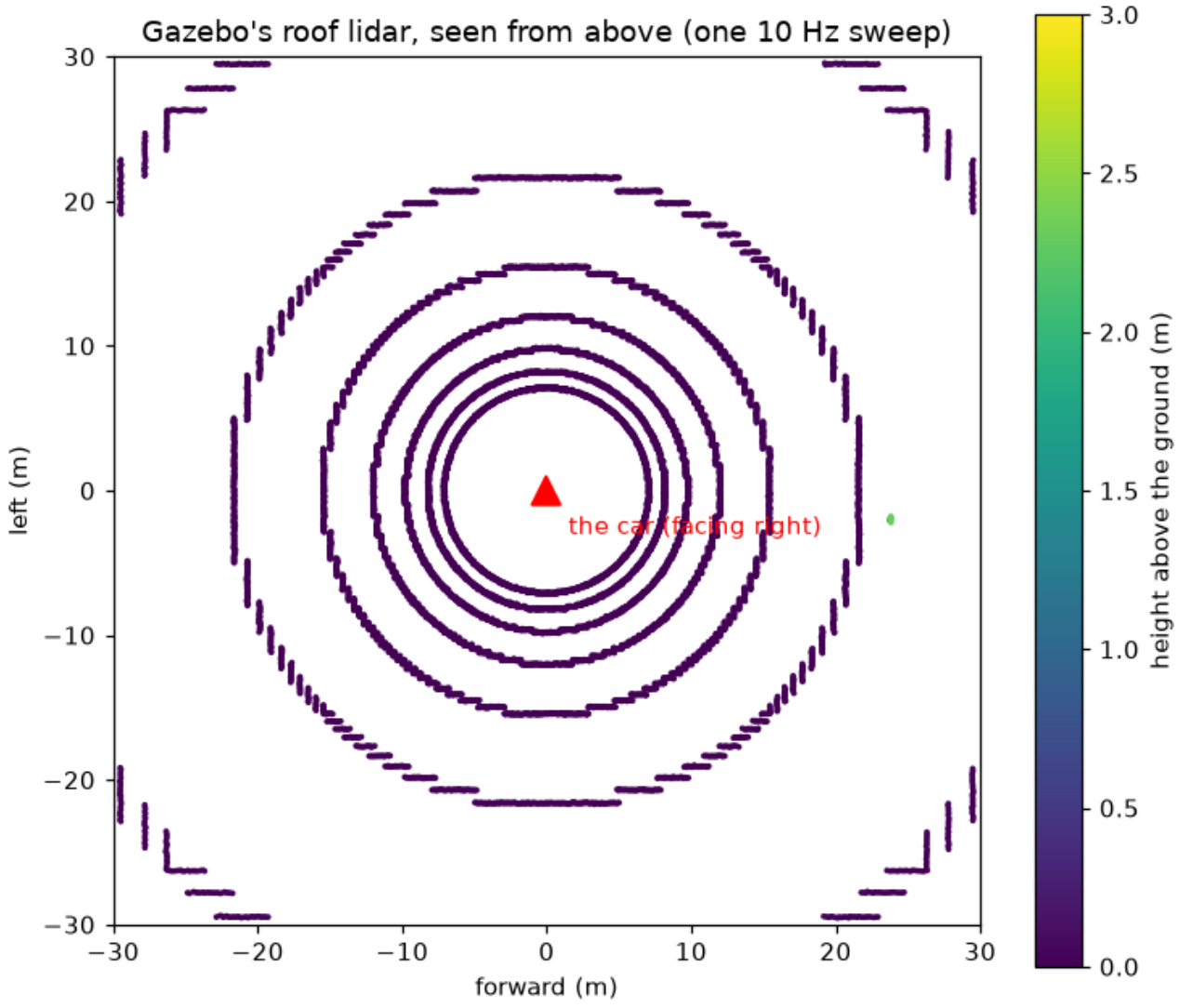


Navigation in 2D and 3D: the lidar for Nav2

The lidar sends about 29,000 points ten times a second. Nav2 reads them one of two ways:

	Flat scan (2D, the default)	Cubes (3D, a trial: `voxels:=true`)
What	one slice: 720 distances round the car	0.25 m cubes, measured from the road fitted to each cloud

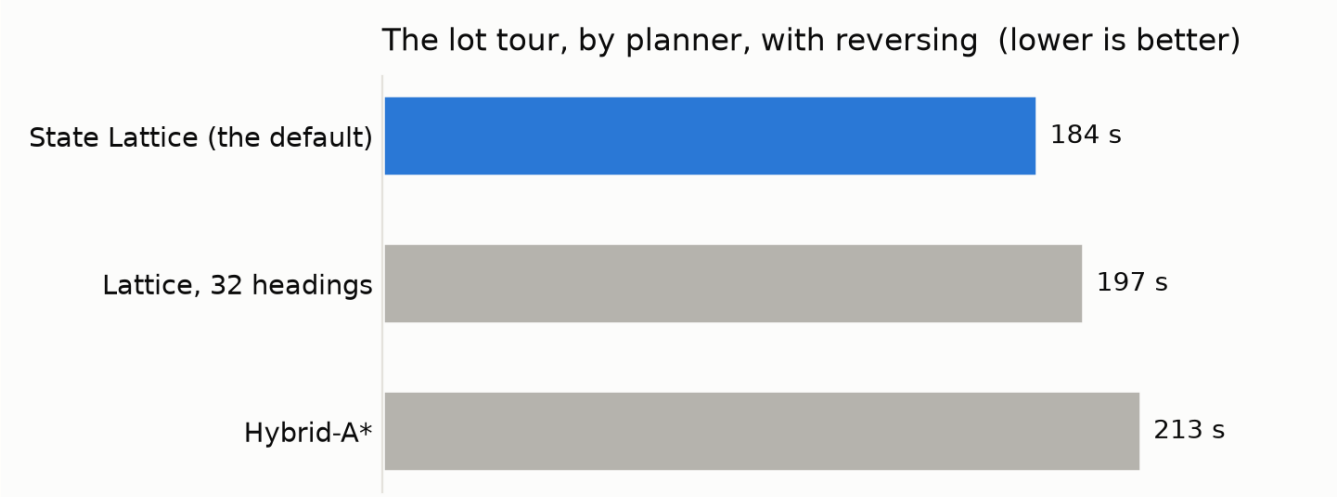
	Flat scan (2D, the default)	Cubes (3D, a trial: `voxels:=true`)
Size	720 numbers	about 1,200 cubes
Sees height?	no	yes: a kerb, a table edge; a leaning car doesn't see the road
Result in Gazebo	the default	stopped 0.44 m before a post; 4 of 4 legs; 6 ms a cloud



Navigation: the route and the steering

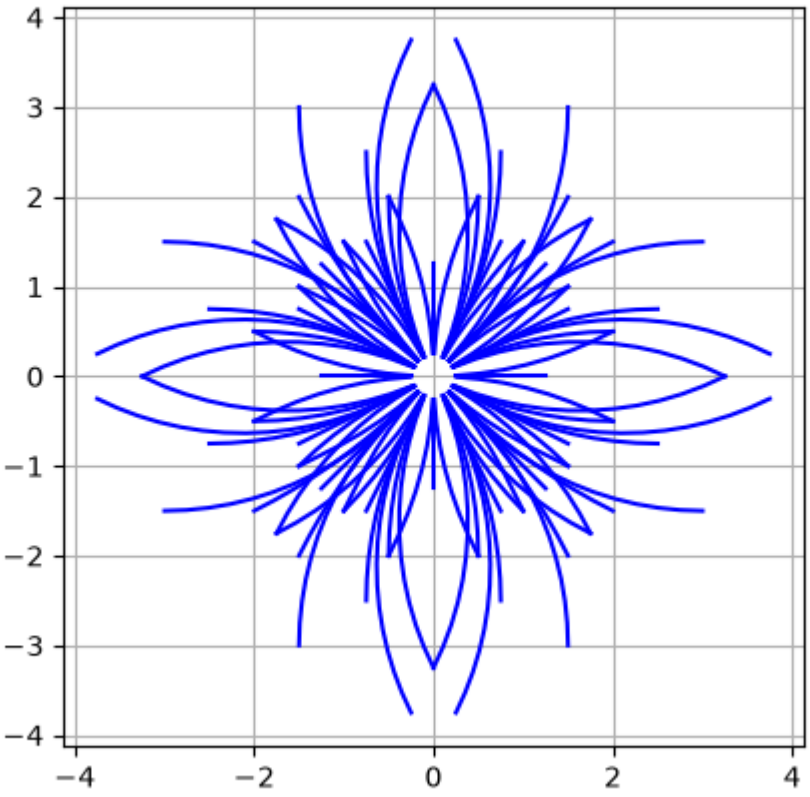
Planners: who draws the route (`planner:=`)

Planner	Keeps to the turning circle?	Gazebo, the lot tour	
State Lattice (on our own moves)	yes	every leg, **184 s** a tour (with reversing)	**the default** since 1 Nov
Hybrid-A*	yes	every leg, 213 s	the default when forward only
Lattice, 32 headings	yes	every leg, 197 s: no better	trial
Smac 2D	no	2 of 4 (forward only)	trial
NavFn, Theta*	no	not tried	trial



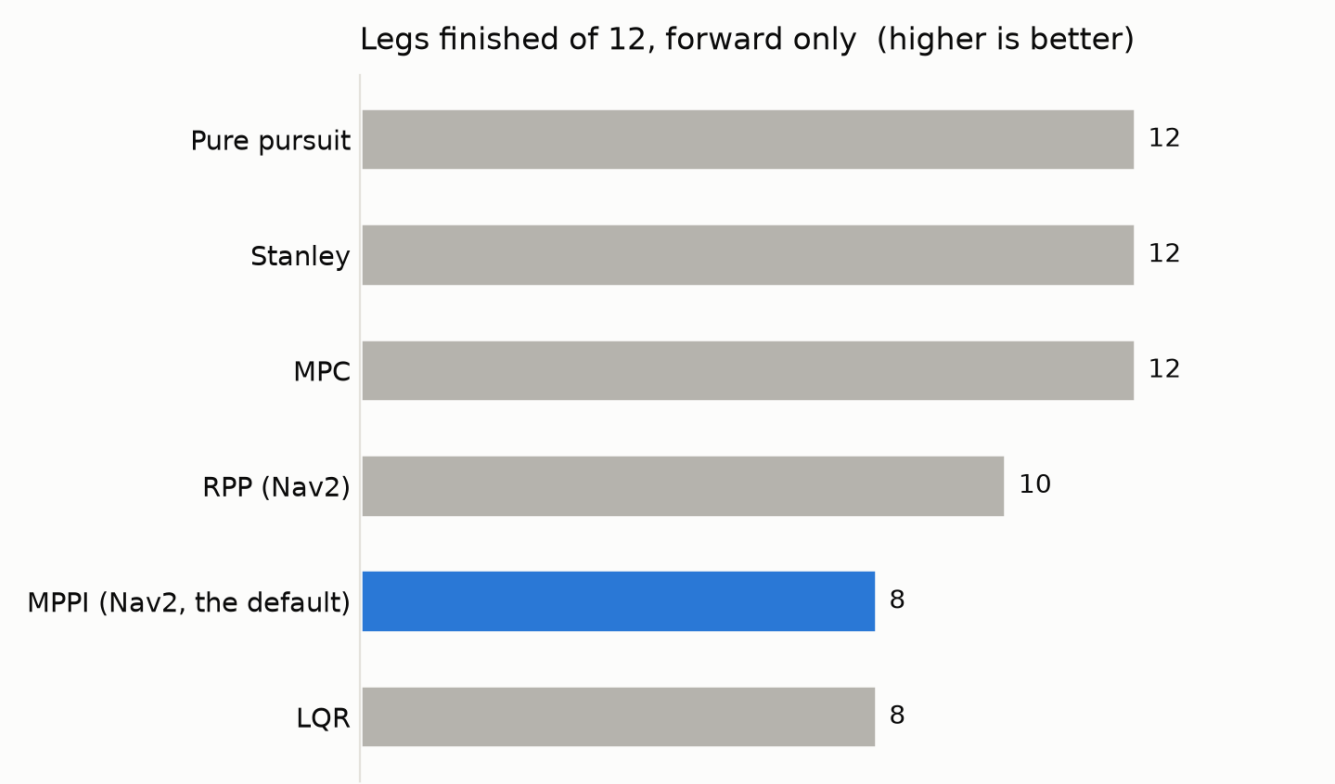
Gazebo, 30 Oct, MPPI; mean of 4 runs (lattice32: 3). Every planner finished every leg.

The lattice strings together short moves our car can really make, worked out from `vehicle.yaml` (`pixi run lattice-primitives`):



Path followers: who steers along it

`tracker:=pure_pursuit` (or `stanley`, `lqr`, `mpc`) swaps them; in Gazebo, live, with keys 1-4 on the debug screen.

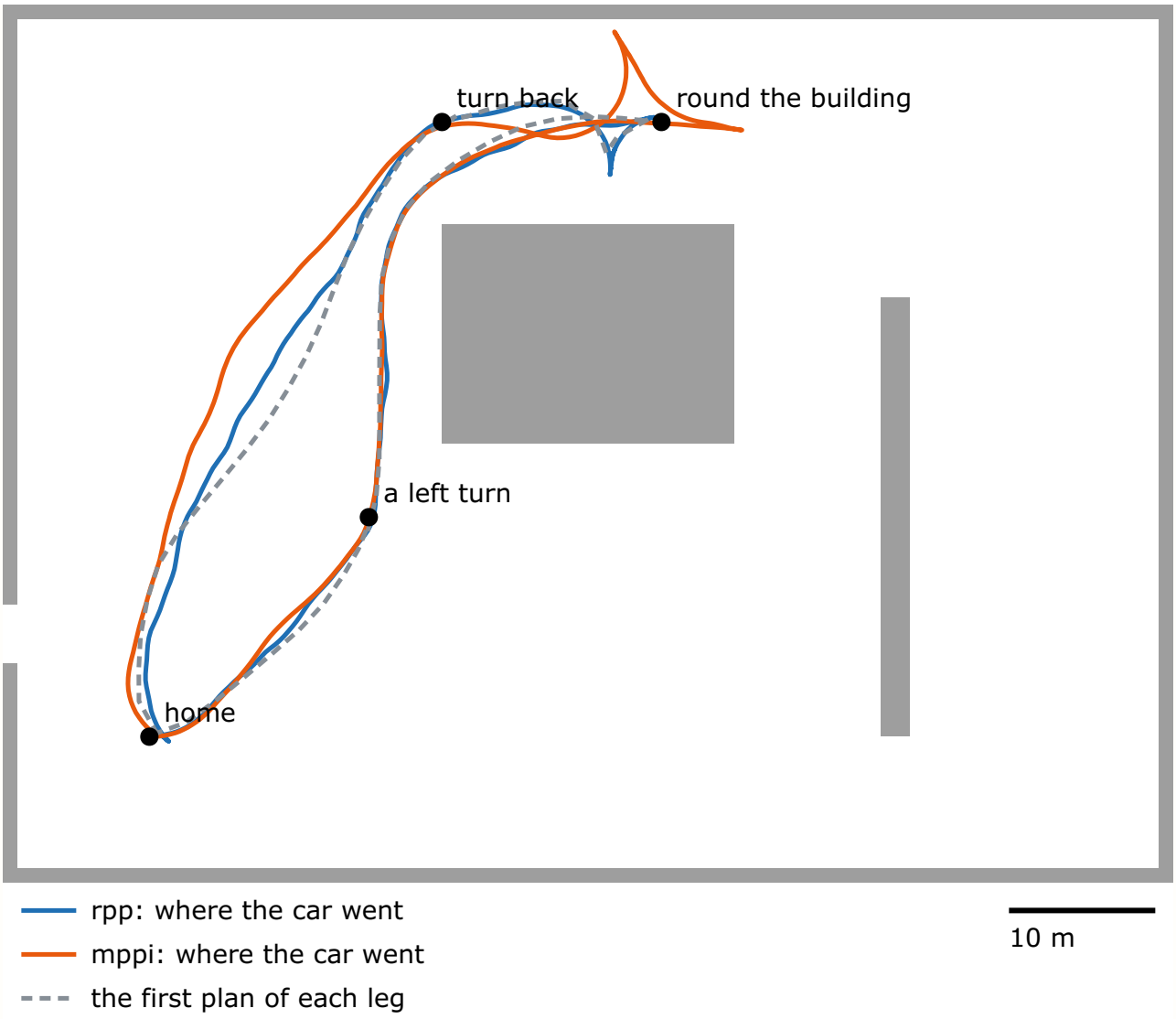


Gazebo's lot, 29 Oct, 3 runs of 4 legs each. The U-turn decided it. MPPI stays the default: it reads the costmap.

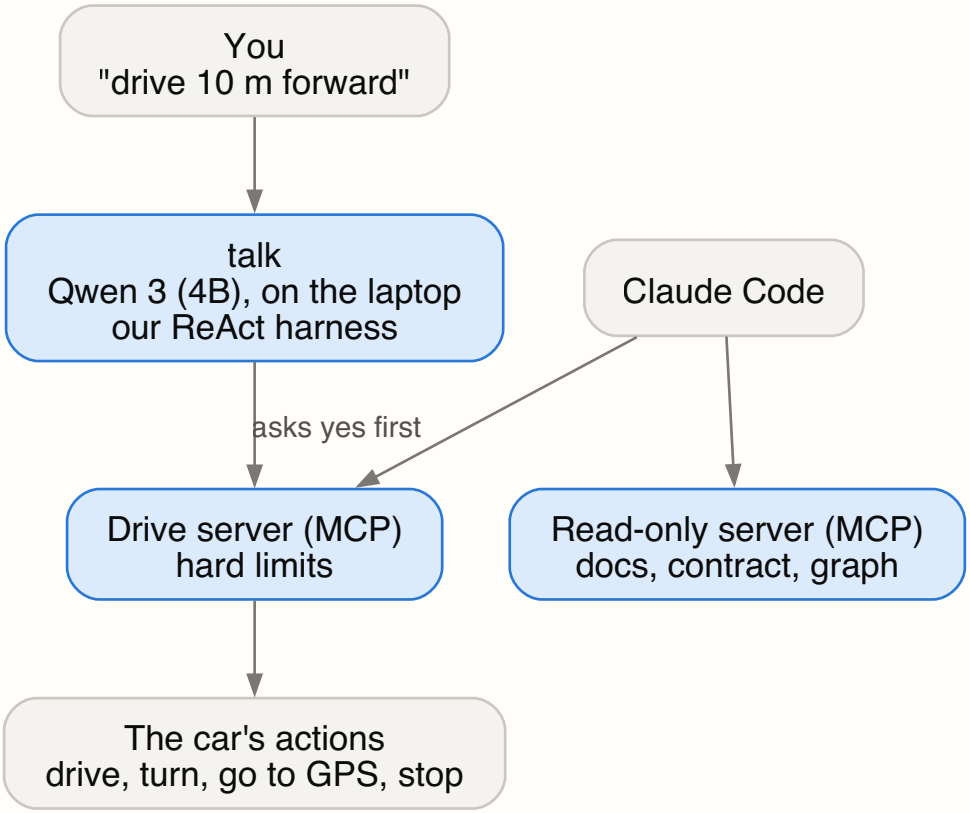
MPPI finished fewest, but stays the default: it reads the costmap, so it steers round obstacles; the trackers only follow the line.

Reversing and the U-turn

On again on the car since 1 Nov, at 0.5 m/s at most, only where the lidar watches behind. Forward only, a U-turn in a narrow lane fails about half the time; reversing, the car makes a three-point turn (16 of 16 legs with the lattice).


turn back
round the building
a left turn
home
rpp: where the car went
mppi: where the car went
the first plan of each leg
10 m

Talking to the car



Piece	What it does	Status
Qwen 3 (4B) with our own harness (`pixi run -e llm talk`)	plain English ("drive 10 m forward") into steps, on the laptop, no internet; asks **yes** before the car moves	sim
The harness (`aav_language`: `react.py`)	our own ReAct loop: reads Qwen's, Llama's, Gemma's and Mistral's tool formats	sim
MCP drive server (`aav-drive`)	lets Claude (or any MCP app) drive: distance, turn, go to a GPS point, stop, e-stop; hard limits it won't break	sim
MCP read-only server (`aav`)	lets Claude read the docs, the topic contract and the live ROS graph; never writes	built

The best version: what goes on the car

One pick per job: the car's defaults. The trials stay as switches; real data decides them next (Every test, 4e: record the lot, then score them on replay).

Job	The best version	Why	Next in line, and what it waits for
Where am I	the two EKFs, with the GPS	works anywhere, no map needed	**NDT into EKF global**: a lidar map of the lot (test 4.21)
How far it moved	wheels + IMU	always there	**KISS-ICP into EKF local**: real wheels that slip (4.22)
What's around	RF-DETR Nano	twice YOLO's range, Apache-2.0	-
Sees obstacles	the flat lidar scan	small, proven	**cubes**: a real, bumpy lot (4.22)
Draws the route	State Lattice (Hybrid-A* forward only)	every leg, the quickest	-
Steers along it	MPPI	reads the costmap	pure pursuit, if the trackers learn the costmap
Reverses?	yes, 0.5 m/s, where the lidar sees behind	the U-turn needs it	wheels that tell direction (H5)
Talks	Qwen on the laptop	no internet, asks yes first	Claude through MCP: the sim only

`test_car_launch.py` checks the car's defaults are exactly these.

CARLA: what's left to work with our graph

CARLA runs on the lab's Ubuntu desktop. Our side (`aav_carla`, `pixi run carla-car`) publishes the same topics as Gazebo, so the rest of the graph can't tell them apart. It is tested against a pretend server only: **it has never met a real one.**

#	Left	Who	Size
1	**Which CARLA** is on the desktop: `CarlaUE4` = 0.9.x, `CarlaUnreal` = 0.10. Its Python client, the same version, in our environment (CARLA, step 2)	the lab	30 min
2	**The first real run:** `pixi run carla-car host:=127.0.0.1` with the server up; drive with the keyboard, then a Nav2 goal	the lab	1 h
3	**A `carla` profile in the graph check** (`pixi run check carla`): the contract doesn't list which topics must be live in CARLA yet	Claude, now	small
4	**The new switches in `carla.launch.py` :** it passes reversing, but not `planner` , `tracker` , `voxels` , `kiss_icp` , `kiss_ekf` (it gets Nav2's defaults)	Claude, now	small
5	**A `carla-check`** , like `sim-check` : the graph, a drive, the EKF against CARLA's truth	Claude, then the lab	1 day
6	**Signs nobody has seen:** the IMU's roll and pitch, the radar's speed sign, the GPS line-up	the lab, with 5	1 h
7	**Carleton's streets:** only CARLA 0.9.16 builds roads from OpenStreetMap; 0.10 can't yet	the lab	1 h
8	**NDT in CARLA:** needs a lidar map of the CARLA town (`ndt-map` drives Gazebo only today)	Claude	half a day

Not on the list, on purpose: CARLA's own ROS 2 (`--ros2`) stalls on Jazzy, so our adapter talks to CARLA through its Python client instead (CARLA, "Other ways").

The computers: which Ubuntu

Can each one be upgraded? Yes: Ubuntu 24.04 (Jazzy's own) runs on the Pi 4 and the Pi 5, and NVIDIA's JetPack 7.2 is Ubuntu 24.04 for the Orin. **But car day doesn't need it:** our car kit runs on today's Ubuntu 22.04 through Pixi (CI tests that on every push). Upgrading is the hardware lead's call (Q24).

Computer	Today	Should be	How	Note
Orin (JETSON, .140)	Ubuntu 22.04, JetPack 6	JetPack 7.2 (Ubuntu 24.04)	SDK Manager on the lab's desktop, **after** its full image	the only one that needs the x86 desktop

Computer	Today	Should be	How	Note
Motor Pi (CORTEX, .120), Pi 4	Ubuntu 22.04	Ubuntu 24.04 Server	a **new SD card** (Raspberry Pi Imager); turn the UART on	keep the old card: undo is a card swap
LOCALIZATION (.113), Pi 4	Ubuntu 22.04	Ubuntu 24.04 Server	a new SD card	the IMU and GPS are USB: no change
aav-desktop (.196), Pi 4, the dashboard	Ubuntu 22.04	**Ubuntu 24.04 Desktop**	a new SD card	runs the HUD and the map window: it needs a desktop; check it has 4 GB or more
`.211`, a Pi	Ubuntu 22.04	ask the team first		its job is unknown
CAM1-PI, **CAM2-PI** (.108, .118), Pi 5	Debian 13 (Pi OS)	**keep Pi OS**	-	no ROS on them; the Pi camera works best on Pi OS
The lab's desktop (x86, NVIDIA)	Ubuntu, running CARLA	**keep as it is**	-	our stack runs there through Pixi on 22.04 or 24.04; don't upgrade under a working CARLA

Tools to find all this again

Tool	Does
`pixi run pick`	a menu of every switch above, in plain words; it runs the command
Switches	every switch on one page
Trials	every result, and how it was measured
`pixi run check <profile>`	the running graph against the contract, then the health line

Tool	Does
`pixi run python3 tools/whats_new_figures.py`	redraws this page's diagrams and charts